

From Theorem 3, this matrix equation has the same solution set as the system of equations corresponding to the augmented matrix

$$\begin{bmatrix} 1 & -2 & 3 & 0 \\ 5 & -9 & h & 0 \\ -3 & 6 & -9 & 0 \end{bmatrix} \xrightarrow[\text{-5} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 3 & 0 \\ 0 & 1 & h-15 & 0 \\ -3 & 6 & -9 & 0 \end{bmatrix} \xrightarrow[\text{+3} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & 3 & 0 \\ 0 & 1 & h-15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & 0 \\ 0 & 1 & h-15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1}]{\text{Row 2} \times 2} \begin{bmatrix} 1 & 0 & 2h-27 & 0 \\ 0 & 1 & h-15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

- Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.
 - Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2
 $\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.
 - Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all posns. below it must contain 0 is True.
 - Each leading entry must be 1 is True.
 - Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all posns. above it must contain 0 is True.
- $\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 2h-27 & 0 \\ 0 & 1 & h-15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in rref. $\square$
- Column 3 has no pivots. By defn, x_3 is a free variable.
 $\therefore$ The system of eqns. has infinitely many solutions.
- By Theorem 2, the system of eqns. has infinitely many solutions for all h $\square$
- $\therefore$ The vectors are linearly dependent for all h $\square$